Nazia Shehnaz Joynab

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Education

- **University of Texas, Dallas** — Dallas, TX
  Master of Science in Computer Science — *Jan 2026 - Aug 2027 (Exp.)*
  - Graduate Researcher @ Dr. Latifur Khan's Lab.
  - Courses Taken: Design and Analysis of Computer Algorithms, Artificial Intelligence, Machine Learning, Robotics
- **Military Institute of Science and Technology** — Dhaka, Bangladesh
  Bachelor of Science in Computer Science and Engineering; CGPA: 3.65/4.00 — *Feb. 2019 – May 2023*

Research Interests

• AI/ML • NLP • Security • Reinforcement Learning • Computer Vision • Robotics

Professional Experience

- **Samsung R&D Institute Bangladesh** — Dhaka, Bangladesh

  ***Software Engineer I (DevOps)**, Full-time, Onsite* — *Dec 2023 - Oct 2025*
  - Contributed to the development and maintenance of an internal content delivery network named ADS (Artifact Deployment System) which leverages advanced caching methods, proxies and cloud technologies to optimize transmission of large binary artifacts across 23 global R&D centers of Samsung Electronics.
  - Engineered an admin portal and CI plugin using Django and J2EE, cutting cloud storage cost by ensuring consistent artifact retention across on-premise and cloud infrastructures (AWS S3).
  - Automated a complex multi-step artifact migration process between storage servers using Bash and Fabric, achieving a **1.75x** speed improvement compared to standard transfer methods.
  - Engaged in comprehensive system monitoring, shouldering critical communication with overseas stakeholders.

  ***Full Stack Development Intern**, Full-time, Onsite* — *Jun 2023 - Dec 2023*
  - Designed and implemented a secure and efficient patent and ideation file management tool using CodeIgniter and MySQL.

Publications

- Munir, M. B., Ononto, A. J., **Joynab, N. S.**, Thuraisingham, B., & Khan, L. (2026). TRIS: A Tri-Layer Retrieval Integrity Sieve Against Knowledge Poisoning. *Findings of the Association for Computational Linguistics: EMNLP 2026*. (link)
- **Joynab, N. S.**, Hossain, S. B. (2026, April 28). From threads to trajectories: A multi-llm pipeline for community knowledge extraction from github issue discussions. arXiv.org. https://arxiv.org/abs/2604.25880
- **Joynab, N. S.**, Islam, M. N., Aliya, R. R., Hasan, A. R., Khan, N. I., & Sarker, I. H. (2024). A federated learning aided system for classifying cervical cancer using pap-smear images. *Informatics in Medicine Unlocked*. (link , pdf , data and materials)

Projects

- **Sim2Real2Sim Robot Sling Puck on a Low-Cost Arm (Ongoing)**
  - Building a **Sim2Real2Sim** pipeline in which a low-cost **SO-101** arm plays sling puck, a fast two-player game, against a simulated human in **MuJoCo**, using scripted grasp, draw, release and block skills.
  - Trained a shot-success model on 4,000 simulated shots (95.4% accuracy) and a **PPO** policy that decides when to shoot or block (70% win rate, on par with a greedy baseline).
  - Built the sim-to-real bridge: arm and board calibration (7.7 mm error), trajectory replay on the real arm, and an **OpenCV** overhead tracker that finds all 20 pucks in recorded video at 8 ms per frame.
  - Designed a failure-guided loop that uses real-world failures (missed grasps, failed shots) to calibrate the simulator and retrain the policy.

- **Can a Poisoned Judge Be Caught? Backdoors in World-Model Policy Evaluators (Ongoing)**
  - Studying whether a few poisoned training clips can make a pretrained world-model judge pass a failing robot policy, while it still ranks honest policies correctly, and whether defenses can catch it.
  - Building the judge: a pretrained **DINO-WM** world model imagines **Meta-World** robot-arm rollouts of about 8 policies per task, and a success head scores them; the ranking is checked against simulator ground truth.
  - Testing 3 public backdoor attacks and 3 defenses, plus our own action-signature trigger, which changes no pixel, and a Physics Cross-Check defense that compares the judge's verdict with the physical state.
  - Analysing how a trigger spreads over the imagined horizon, where the backdoor is stored (activation patching), and how much poison it takes to flip a ranking.
- **Federated learning aided cervical cancer classification (Undergrad Thesis)**
  - Proposed a CNN-based federated learning architecture for cervical cancer classification which showed test accuracy of **94.36%** and **78.4%** on an IID (Independent and Identically Distributed) and a non-IID setting respectively.
- **TRIS: Tri-Layer Retrieval Integrity Sieve - Big Data Management and Analytics Lab, UTD**
  - Contributed to TRIS, a tri-layer middleware defense for mitigating knowledge poisoning attacks in Retrieval-Augmented Generation (RAG) systems.
  - Implemented and evaluated retrieval integrity mechanisms using semantic clustering, structural trigger–payload filtering, and LLM consistency verification.
  - Conducted adversarial evaluations across NQ, HotpotQA, and MS-MARCO, achieving attack success rates as low as 3% on NQ and 4% on MS-MARCO in the evaluated black-box setting.
- **Joyita: An ML based screening application for facilitating the cervical cancer treatment**
  - Developed an android application with re-usable widgets in **Flutter**. Used **Cloud Firestore** to store doctor's appointment details, patient blogs etc. The ML model predicted the possibility of having cervical cancer based on user inputs of a pre-defined questionnaire displayed on the app's UI. Hosted ML model using **Flask and Heroku**.
  - Testing of the application was done by almost **50+** users.

Programming Skills

- **Programming Languages**: Python, Java, Bash, C++, SQL, JavaScript
- **ML Frameworks/Libraries**: Pandas, Scikit-learn, Keras, Tensorflow, Pytorch, OpenCV, LangChain, Agno
- **Robotics & Simulation:** ROS 2, NVIDIA Isaac Sim
- **Backend Frameworks**: Django, Laravel, CodeIgniter, Flask
- **Databases**: MySQL, Postgresql, ChromaDB, Oracle, Firebase
- **Android**: Dart, Flutter
- **DevOps**: Linux, Docker, Kubernetes (Basic), AWS S3 (Basic), CI/CD, GitHub, Splunk

Achievements

- **Tech Innovator**, MX SE Award for ADS cloud mode development, Samsung Electronics (2025) [link]
- **2nd Runners-up**, Devmania by BDOSN (2022): Developed AlertTrace, a wearable SOS system with real-time location tracking and emergency alerts.

Problem Solving Profile

Leetcode , **Hackerrank** , **Vjudge** , **CSES** (Solved **600 problems** across all platforms.)